

Chapter 8

Monads

The notion of monads originates in homological algebra and category theory.

8.1 Definition

Here are some notations for defining monads.¹

- Any endofunctor $T : \mathcal{C} \longrightarrow \mathcal{C}$ has composites $T^2 = T \circ T : \mathcal{C} \longrightarrow \mathcal{C}$ and $T^3 = T^2 \circ T : \mathcal{C} \longrightarrow \mathcal{C}$.
- If $\mu : T^2 \rightrightarrows T$ is a natural transformation, with components $\mu_x : T^2x \longrightarrow Tx$ for each $x \in \mathcal{X}$, then
 - $T\mu : T^3 \longrightarrow T^2$ denotes the natural transformation with components $(T\mu)_x = T(\mu_x) : T^3x \longrightarrow T^2x$.
 - $\mu T : T^3 \longrightarrow T^2$ denotes the natural transformation with components $(\mu T)_x = \mu_{Tx} : T^3x \longrightarrow T^2x$.

¹MacLane (1997, p.137)

Definition 8.1.1 (Monad²). A *monad* $T = \langle T, \eta, \mu \rangle$ in $\mathcal{C}$ consists of a functor $T : \mathcal{C} \rightarrow \mathcal{C}$ and two natural transformations

$$\eta : id_{\mathcal{C}} \xrightarrow{\sim} T, \quad \mu : T^2 \xrightarrow{\sim} T$$

which make the following diagrams commute for any object A in $\mathcal{C}$:

$$\begin{array}{ccc} T^3 A & \xrightarrow{T\mu_A} & T^2 A \\ \mu_{TA} \downarrow & & \downarrow \mu_A \\ T^2 A & \xrightarrow{\mu_A} & TA \end{array} \quad \begin{array}{ccc} TA & \xrightarrow{\eta_{TA}} & T^2 A \xleftarrow{T\eta_A} TA \\ id_{\mathcal{C}} \searrow & & \downarrow \mu_A \swarrow id_{\mathcal{C}} \\ & TA & \end{array}$$

We call η the *unit* and μ the *multiplication* or the *join*.

The left diagram is *associativity law*, and the right diagram is *unitary law* where η plays a role of unit for multiplication μ .

Theorem 8.1.2. For any adjunction $\langle F, G, \eta, \epsilon \rangle : \mathcal{X} \rightarrow \mathcal{A}$,

$$\langle GF, \eta : id_{\mathcal{X}} \rightarrow T, G\epsilon_{F-} : T^2 \rightarrow T \rangle$$

is a monad. This is called a *monad defined by the adjunction*.

Proof. We show that the triple defined by a given adjunction as above qualifies as monad. Regarding the associativity law:

$$\begin{aligned} \mu_A \circ T\mu_A &= G\epsilon_{FA} \circ GF G\epsilon_{FA} \\ &= G(\epsilon_{FA} \circ FG\epsilon_{FA}) \\ &= G(\epsilon_{FA} \circ \epsilon_{FGFA}) \quad (\text{by naturality of } \epsilon) \\ &= G\epsilon_{FA} \circ G\epsilon_{FGFA} \\ &= \mu_A \circ \mu_{TA} \end{aligned}$$

The unitary laws are proved as below by using the triangular identity of adjunctions:

$$\begin{aligned} \mu_A \circ \eta_{TA} &= G\epsilon_{FA} \circ \eta_{GFA} = id_{GFA} \\ \mu_A \circ T\eta_A &= G\epsilon_{FA} \circ GF\eta_A = G(\epsilon_{FA} \circ F\eta_A) = G(id_{FA}) = id_{GFA} \end{aligned} \quad \square$$

²MacLane (1997, p.137:(1)(2))

Example 8.1.3. Recall the adjunction $_ \times A \dashv _{}^A$ where $\eta = \lambda(id_{_ \times A})$ and $\epsilon = ev_{A, _}$. This adjunction defines a state monad.

8.2 T -algebra

T -algebra is the construction due to ?.

Definition 8.2.1 (T -algebra). For any monad $\langle T, \eta, \mu \rangle$ in a category $\mathcal{X}$, T -algebra $\langle X, h \rangle$ is a pair consisting of an object $X \in ob\mathcal{X}$ and a morphism $h : TX \rightarrow X$ that makes the following diagrams commute:

$$\begin{array}{ccc} T^2X & \xrightarrow{Th} & TX \\ \mu_X \downarrow & & \downarrow h \\ TX & \xrightarrow{h} & X \end{array} \qquad \begin{array}{ccc} X & \xrightarrow{\eta_X} & TX \\ & \searrow id_X & \downarrow h \\ & & X \end{array}$$

X and h are called *the underlying object* and *the structure map* of the algebra.

The left and right diagrams respectively represents the associative law and the unit law of the algebra.

Definition 8.2.2 (Category of T -algebra). For any monad $\langle T, \eta, \mu \rangle$ in a category $\mathcal{X}$, a category of T -algebra (notation X^T) consists of (as objects) all T -algebras, and (as morphisms from $\langle X, h \rangle$ to $\langle X', h' \rangle$) $\mathcal{X}$ -morphisms of the form $f : X \rightarrow X'$ that makes the following diagram commute:

$$\begin{array}{ccc} X & \xleftarrow{h} & TX \\ f \downarrow & & \downarrow Tf \\ X' & \xleftarrow{h'} & TX' \end{array}$$

Identity arrows and compositions are defined in a straightforward manner.

The following theorem shows a decomposition of a given monad into adjunction.

Theorem 8.2.3. For any monad $\langle T, \eta, \mu \rangle$ in $\mathcal{X}$, there is an adjunction $\langle F^T, G^T, \eta^T, \epsilon^T \rangle : \mathcal{X} \longrightarrow \mathcal{X}^T$ where

- $F^T : \mathcal{X} \longrightarrow \mathcal{X}^T$ sends X to $\langle TX, \mu_X \rangle$ and f to Tf .
- $G^T : \mathcal{X}^T \longrightarrow \mathcal{X}$ sends $\langle X, h \rangle$ to X and f to f .
- η^T is η .
- ϵ^T sends $\langle X, h \rangle$ to h .

Proof.

□

Theorem 8.2.4. The adjunction $\langle F^T, G^T, \eta^T, \epsilon^T \rangle$ defines the original monad $\langle T, \eta, \mu \rangle$.

Proof.

□

8.3 Kleisli Category

Kleisli category is the construction due to Kleisli (1965).

Definition 8.3.1 (Kleisli category). For any monad $\langle T, \eta, \mu \rangle$ in $\mathcal{X}$, the *Kleisli category* $\mathcal{X}_T$ consists of (as objects) X_T in $\mathcal{X}_T$ for each object X in $\mathcal{X}$, and (as morphism) $f^\flat : X_T \longrightarrow Y_T$ in $\mathcal{X}_T$ for each morphism $f : X \longrightarrow TY$ in $\mathcal{X}$.

The composite of $f^\flat : X_T \longrightarrow Y_T$ with $g^\flat : Y_T \longrightarrow Z_T$ is defined by $g^\flat \circ f^\flat = (\eta_Z \circ Tg \circ f)^\flat$.

Theorem 8.3.2. For any monad $\langle T, \eta, \mu \rangle$ and a Kleisli category $\mathcal{K}_T$, functors $F_T : \mathcal{K} \rightarrow \mathcal{K}_T$ and $G_T : \mathcal{K}_T \rightarrow \mathcal{K}$ are defined by

$$\begin{aligned} F_T : k : X \rightarrow Y &\mapsto (\eta_y \circ k)^\flat : X_T \rightarrow Y_T \\ G_T : f^\flat : X_T \rightarrow Y_T &\mapsto \eta_Y \circ Tf : TX \rightarrow T^2Y \rightarrow TY \end{aligned}$$

respectively, so that $G_TX_T = TX$ on objects.

There is an adjunction $\langle F_T, G_T, \eta_T, \varepsilon_T \rangle$ where $\eta_T = \eta$ and $\varepsilon_T = \eta$.

Proof. □

Theorem 8.3.3. The adjunction $\langle F_T, G_T, \eta_T, \varepsilon_T \rangle$ defines the original monad $\langle T, \eta, \mu \rangle$.

Proof. □

8.4 Comparison Theorem

Theorem 8.4.1 (Comparison Theorem for T -algebra). Given an adjunction $\langle F, G, \eta, \varepsilon \rangle : \mathcal{K} \rightarrow \mathcal{A}$ and $\langle GF, \eta, G\varepsilon_F _ \rangle$ is a monad that it defines in $\mathcal{K}$, there is a unique functor $K : \mathcal{A} \rightarrow \mathcal{K}^T$ with $G^TK = G$ and $KF = F^T$.

Proof. □

Theorem 8.4.2 (Comparison Theorem for Kleisli Category). Given an adjunction $\langle F, G, \eta, \varepsilon \rangle : \mathcal{K} \rightarrow \mathcal{A}$ and $\langle GF, \eta, G\varepsilon_F _ \rangle$ is a monad that it defines in $\mathcal{K}$, there is a unique functor $L : \mathcal{K}^T \rightarrow \mathcal{A}$ with $GL = G_T$ and $LF_T = F$.

Proof. □

Definition 8.4.3. Given a monad $\langle T, \eta, \mu \rangle$, there is a category with objects all adjunctions $\langle \mathbf{F}, \mathbf{G}, \eta, \varepsilon \rangle : \mathcal{X} \longrightarrow \mathcal{A}$ which defines $\langle T, \eta, \mu \rangle$ in $\mathcal{X}$, and with morphisms maps of adjunctions.

Theorem 8.4.4. The above category has an initial object $\langle \mathbf{F}_T, \mathbf{G}_T, \eta_T, \varepsilon_T \rangle$ and a terminal object $\langle \mathbf{F}^T, \mathbf{G}^T, \eta^T, \varepsilon^T \rangle$ with the comparison functor:

$$\mathcal{X}_T \xrightarrow{L} \mathcal{A} \xrightarrow{K} \mathcal{X}^T$$

8.5 Kleisli Triple

There is an alternative description of a monad (cf. Moggi (1989)). The Kleisli triples is introduced by E. Manes in his 1976 book called "Algebraic Theories".

Definition 8.5.1 (Kleisli triples). ³A Kleisli triple over $\mathcal{X}$ is a system(?) consisting of functions:

- (i) $T: ob\mathcal{X} \longrightarrow ob\mathcal{X}$
- (ii) $\eta: ob\mathcal{X} \longrightarrow mor\mathcal{X}$ with value $\eta_A: A \longrightarrow TA$ on object A
- (iii) $-*: a$ function on arrows $f: A \longrightarrow TB$ with value $f^*: TA \longrightarrow TB$

which is subject to the following equations:

$$(K1) \quad (\eta_A)^* = id_{TA}$$

$$(K2) \quad f^* \circ \eta_A = f$$

$$(K3) \quad g^* \circ f^* = (g^* \circ f)^* \text{ for } g: B \longrightarrow TC$$

³Pedicchio and Tholen (2004, p.249-)

Theorem 8.5.2. Every Kleisli triple $\langle T, \eta, -^* \rangle$ over $\mathcal{X}$ has a unique extension to a monad $\langle T, \eta, \mu \rangle$ on $\mathcal{X}$ with:

- $Th = (\eta_B \circ h)^*$ for each morphism $h : A \rightarrow B$ of $\mathcal{X}$
- $\eta_A = (id_{TA})^*$

Conversely, every monad on $\mathcal{X}$ restricts to a Kleisli triple with $f^* = \eta_B \circ Tf$ for each $f : A \rightarrow TB$ in $\mathcal{X}$.

These processes are inverse to each other.

Exercise 8.5.3. Prove Theorem 8.5.2.

Theorem 8.5.4. The morphism f is an isomorphism in $\mathcal{X}_T$ iff f^* is an isomorphism in $\mathcal{X}$.

Exercise 8.5.5. Prove Theorem 8.5.4.

Theorem 8.5.6. Given diagrams

$$\begin{array}{ccc} A & \xrightarrow{f} & B \\ h \downarrow & & \downarrow g \\ C & \xrightarrow{k} & D \end{array} \quad \begin{array}{ccc} TA & \xrightarrow{f^*} & TB \\ h^* \downarrow & & \downarrow g^* \\ TC & \xrightarrow{k^*} & TD \end{array}$$

it then follows that:

- (1) the left square commutes in $\mathcal{X}_T$ iff the right square commutes in $\mathcal{X}$.
- (2) the left square is a pullback in $\mathcal{X}_T$ iff the right square is a pullback in $\mathcal{X}$.

Exercise 8.5.7. Prove Theorem 8.5.6.

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